Simple Number Pattern Generator

1. Problem Statement

Read the number of rows and print a simple increasing star pattern using nested loops.

2. Objectives

1. Read the number of rows.
2. Use nested `for` loops.
3. Generate a star pattern.
4. Display the pattern row by row.

3. Assumptions

1. The number of rows must be a positive integer.
2. Each row contains stars equal to its row number.

4. Algorithm

1. Start.
2. Read the number of rows.
3. Repeat row from 1 to the number of rows.
4. For each row, print stars equal to the row number.
5. Move to the next line after each row.
6. Stop.

5. Flowchart

None

6. Pseudocode

None

```
INPUT rows

IF rows <= 0 THEN
    DISPLAY "Enter a positive number"
```

```
ELSE
    FOR i = 1 TO rows
        FOR j = 1 TO i
            DISPLAY "*" WITHOUT NEW LINE
        END FOR
        DISPLAY NEW LINE
    END FOR
END IF
```

8. Test Case and Expected Output

Input:

```
None
Enter number of rows: 4
```

Output:

```
None
----- STAR PATTERN -----
*
**
***
****
```

Explanation:

- Row 1 contains 1 star.
- Row 2 contains 2 stars.
- Row 3 contains 3 stars.
- Row 4 contains 4 stars.

9. Conclusion

The program generates an increasing star pattern using nested `for` loops. Each row displays stars equal to its row number.